TARSO HEBERT CAMILO VIEIRA

Software Engineer | Python · Backend · Automation

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SUMMARY

Software engineer at TRF1, a Brazilian federal court, where I build and maintain the automation suite that drives the electronic case system: **40 bots in production**, **370 internal users** and over **1.9 million case files** analyzed with a **97.32% success rate** across 15,000 runs. I work end to end — requirements with non-technical staff, Python development, automated testing and production support — and I also build web applications with Flask and Django. Computer Science undergraduate, graduating in 2027.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C

Backend & APIs: Flask, Django, Django REST Framework, SQLAlchemy, Node.js, Express, REST APIs, system integration

Automation: Selenium WebDriver, RPA, web scraping, process automation, PyInstaller, pandas, openpyxl

Testing & Quality: pytest, unit and integration tests, coverage, SonarQube, static analysis, code review

Applied AI: LLM integration in production (Google Gemini), AI classifiers for legal documents

Databases: PostgreSQL, MySQL, SQLite, Redis, relational modeling

DevOps & Practices: AWS (cloud and security fundamentals), Git, GitLab CI, GitHub, Docker, Docker Compose, Nginx, Keycloak, Linux, semantic versioning, Conventional Commits, Scrum, Kanban

PROFESSIONAL EXPERIENCE

Software Engineer

G&E Serviços Terceirizados — contractor at TRF1, Federal Court of the 1st Region | Brasília, Brazil

May 2026 – Present

- Build and maintain the court's automation suite — **40 bots in production**, **370 users** and over **1.9 million case files** analyzed, with a **97.32% success rate** across 15,000 runs.
- Develop bots in Python and Selenium that operate PJe, the electronic case system, across **13 states** and the Federal District, working in **17 repositories** from requirements gathering to production support.
- Created the pytest suites and SonarQube analysis for **20+ bots**, totaling **800+ tests** — **100% coverage** on the vote-registration bot (299 tests) and **99%** on the expert-payment bot (115 tests).
- Refactored critical bots with state-based synchronization, per-item failure isolation and response validation, cutting intermittent failures from **~50% to ~9%** of runs.
- Integrated bots with Google Gemini (LLM) for document analysis and reminder generation, reducing triage time from **20 minutes to 1 minute** (~95% reduction).
- Built a consolidated case-lookup bot for both court instances: isolated parallel sessions and **a single Excel report** with status, court unit and last-update date per case.
- Develop RPAHub, a Flask and SQLAlchemy application with authentication that catalogs and distributes the **40 bots** to internal teams, containerized with Docker and served by Nginx.
- Maintain the shared RPA library reused by **50+ bots** and package deliverables as executables with PyInstaller.
- Gather requirements directly with court staff and non-technical teams, turning legal routines into technical specifications, and mentor one intern.

IT Intern — Automation

CIEE | placed at TRF1, Federal Court of the 1st Region — Brasília, Brazil

Mar 2026 – Apr 2026

- Joined the automation team supporting bot development in Python and Selenium; **hired as a full-time engineer after two months**.

Administrative Apprentice — Data & Operations
Grupo Petrópolis (brewing company) | Bom Jesus da Lapa, Brazil
Jan 2022 – Sep 2023 · 1 yr 9 mos

- Supported logistics data analysis and production planning and control at a large industrial plant, taking part in inventory automation and traceability initiatives.

PROJECTS

APROVA — Exam Preparation Platform — **github.com/Tarso333/ConcursoSedesDF**
Electron · React · TypeScript · SQLite · Drizzle ORM · Tailwind CSS
Offline-first desktop app for Brazilian civil-service exams: mock exams in the official format, spaced-repetition flashcards (FSRS), a PDF syllabus importer and an AI tutor with pluggable providers (local Ollama by default). **84 tests** cover the business-rule engines.

PropostaJá — Quoting App for Independent Workers — **github.com/Tarso333/propostaja**
React Native · Expo · TypeScript · SQLite
Mobile app that lets service providers build a PDF quote and send it over WhatsApp while still at the client's place. Runs **100% offline**, with versioned database migrations and amounts stored in integer cents to avoid rounding errors.

API Monitor — Observability Dashboard — **api-monitor-alpha.vercel.app**
React · Vite · Node.js · Express · Docker Compose · Dynatrace
Public-API monitoring dashboard built by a **team of 9** during the Porto Digital tech residency. I owned the backend and observability: Express endpoints, a Docker Compose environment and latency, throughput and availability metrics in Dynatrace.

Personal Portfolio — **tarsohebert.dev.br**
Astro · TypeScript · Tailwind CSS · Vercel
Static site with an editorial layout, light and dark themes, SEO and accessibility work, deployed on a custom domain with HTTPS and DNSSEC. Scores **100** on Lighthouse performance, accessibility, best practices and SEO.

EDUCATION

B.Sc. in Computer Science — Universidade Católica de Brasília (UCB)
6th semester · Expected graduation: 2027

CERTIFICATIONS & LANGUAGES

Certifications: AWS Academy Graduate – Cloud Foundations (AWS, Oct 2026) · AWS Academy Graduate – Cloud Security Foundations (AWS, Oct 2026) · Docker: Essential Tool for Developers (Udemy, Aug 2026) · Selenium + Python for Web Scraping and Automation (Udemy, Jun 2026) · TakeOff Program: AI, Prompt Engineering, DevOps and Docker (Rocketseat, Apr 2026) · Python Programming from Zero to Artificial Intelligence (Udemy, Apr 2026) · JavaScript from Basic to Advanced with Node.js (Udemy, Feb 2026)
Languages: Portuguese (native) · English (advanced technical reading and writing)